

a Parking Function?

J. Carlos Martínez Mori

Consider a one-way street with $n \in \{1, 2, \dots\}$ numbered parking spots, denoted $s_1, s_2, \dots, s_n$. A sequence of n cars enters the street one at a time, each with a preferred spot. Upon its arrival, car $1 \leq i \leq n$, denoted c_i , drives to its preferred spot $x_i \in [n] := \{1, 2, \dots, n\}$. If spot x_i is unoccupied, car c_i is *lucky* and parks there. Otherwise, car c_i *displaces* further down the street until it finds the first unoccupied spot in which to park, if such a spot exists. If no such spot exists, car c_i reneges the search process unable to park. Let $\mathbf{x} = (x_1, x_2, \dots, x_n) \in [n]^n$ be the n -tuple encoding the cars' parking preferences. If all cars are able to park, then $\mathbf{x}$ is said to be a *parking function* of length n .

Parking functions were first implicitly studied by Pyke [Pyk59] in his study of Poisson processes and later on by Konheim and Weiss [KW66] in their study of hashing with linear probing. Parking functions can be seen as functions in the sense that the "parking experiment" narrated above assigns, to each n -tuple of parking preferences, a single n -tuple encoding its parking *outcome*. If the n -tuple of preferences is indeed a parking function, its outcome can be treated as a permutation of $[n]$ written in one-line notation. For example, as depicted in Figure 1, $(1, 3, 1, 1)$ is a parking function of length 4 with outcome $(1, 3, 2, 4)$.

Classical enumerative results about parking functions foreshadow their rich mathematical structure. There are

$$(n+1)^{n-1} \quad (1)$$

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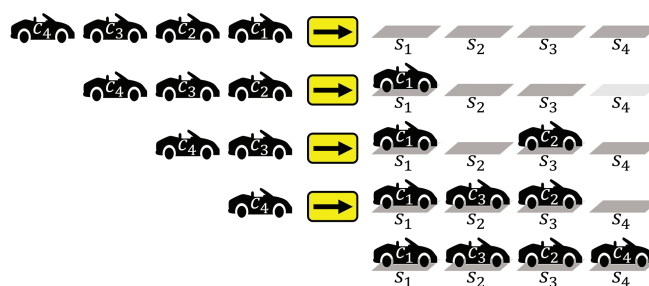
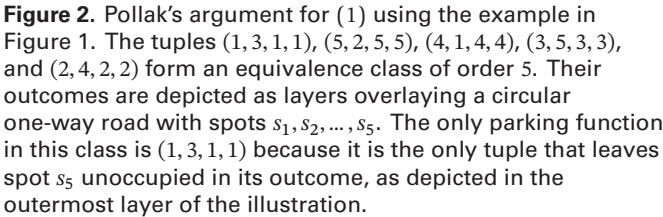


Figure 1. The parking function $(1, 3, 1, 1)$ has outcome $(1, 3, 2, 4)$. Cars c_1 and c_2 are lucky and park in their preferred spots s_1 and s_3 , whereas cars c_3 and c_4 displace further down the street before ultimately parking in spots s_2 and s_4 , respectively.

parking functions of length $n \geq 1$ (OEIS A000272)—this is Cayley's formula [Cay89] for the number of labeled trees on $n+1$ vertices. This count was established independently by Pyke [Pyk59] and Konheim and Weiss [KW66], and has been recovered bijectively by numerous authors; refer to Yan [Yan15, Section 13.2] and the references therein.

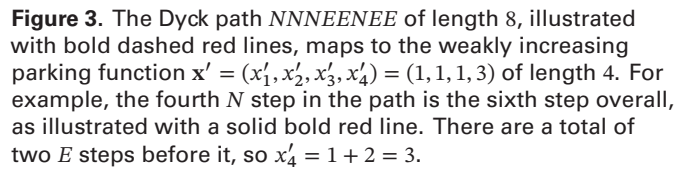
An elegant proof of (1), due to Pollak as credited by Riordan [Rio69], is as follows. Consider a circular one-way street with $n+1$ numbered parking spots, denoted $s_1, s_2, \dots, s_n, s_{n+1}$. Say, it is a roundabout with no exits and a single entry between spots s_{n+1} and s_1 . There are n cars attempting to park, so that each n -tuple $\mathbf{x} = (x_1, x_2, \dots, x_n) \in [n+1]^n$ encodes a possibility for the cars' parking preferences. How many of these n -tuples are in fact parking functions of length n ? Since there are $n+1$ spots but only n cars, there will always be one spot left unoccupied; which exact spot this is depends on the preferences. Note that the preference tuples form a group G under component-wise addition modulo $n+1$, and consider the subgroup $H \leq G$ generated by the all-ones preference tuple $(1, 1, \dots, 1)$. As depicted in Figure 2, the cosets $\mathbf{x}H = \{\mathbf{x} + \mathbf{h} : \mathbf{h} \in H\}$ for $\mathbf{x} \in G$ form equivalence classes of order $n+1$, each of which contains exactly one parking function, namely



whichever tuple leaves spot $n + 1$ unoccupied in its outcome. Therefore, there are $(n + 1)^n / (n + 1) = (n + 1)^{n-1}$ parking functions of length n .

$$\mathbf{x} \text{ is a parking function of length } n \iff x'_i \leq i \text{ for all } 1 \leq i \leq n. \quad (2)$$

Note that (2) has the following remarkable implication: parking functions are invariant under the action of the symmetric group $\mathfrak{S}_n$, which permutes their subscripts. In particular, the set of parking functions is obtained from the rearrangements of weakly increasing parking functions. For example, $(1, 3, 1, 1)$, $(1, 1, 3, 1)$, $(1, 1, 1, 3)$, and $(3, 1, 1, 1)$ are all parking functions of length 4, each with a different parking outcome, whereas no rearrangement of $(1, 3, 3, 4)$ can possibly be a parking function of length 4.



Parking functions are also closely related to the Catalan numbers (OEIS A000108), which are given by the recurrence relation

for $n \geq 1$ with $C_0 = 1$. In particular, the set of weakly increasing parking functions of length n is enumerated by (3). To verify this, suppose you construct a weakly increasing parking function of length n , denoted $\mathbf{x}' = (x'_1, x'_2, \dots, x'_n) \in [n]^n$, with your choice of $1 \leq i \leq n$ for the greatest index satisfying $x'_i = i$. For any such choice of i , the $(i-1)$ -tuple $(x'_1, x'_2, \dots, x'_{i-1})$ must be a parking function of length $i-1$, the $(n-i)$ -tuple $(x'_{i+1} - i + 1, x'_{i+2} - i + 1, \dots, x'_n - i + 1)$ must be a parking function of length $n-i$, and by induction there are $C_{i-1} \cdot C_{n-i}$ possibilities.

Note that if $\mathbf{x}' = (x'_1, x'_2, \dots, x'_n) \in [n]^n$ is a weakly increasing parking function, then the i th car c_i (with preference x'_i) parks in the i th spot s_i . Therefore, the total

displacement of a weakly increasing parking function $\mathbf{x}'$ is given by

$$\sum_{i=1}^n i - x'_i = \frac{n(n+1)}{2} - \sum_{i=1}^n x'_i. \quad (4)$$

This statistic has further combinatorial interpretations. For example, the number of full squares between a Dyck path and the main diagonal, which in the case of Figure 3 is 4, is the same as the total displacement of its corresponding weakly increasing parking function. Lastly, note that the dependency of (4) on $\mathbf{x}'$ reduces to the summation of its terms. This implies that the total displacement of a parking function is also preserved under rearrangements!

Despite first appearing in the literature more than six decades ago, the combinatorics of parking functions remains a vibrant area of research. Recent work ranges from the study of discrete statistics such as displacement [KY23, EHKMM23] or the number of lucky cars [GS04, SV24, SY23], the introduction of new variants and/or generalizations on the classical “parking experiment” rule (refer to Carlson et al. [CCH⁺21] for an accessible tour of endless possibilities in the style of “choose your own adventure”), polyhedral aspects [AW22, HLVM24], and surprising connections to seemingly unrelated objects [AAH⁺23, HKMM24], to mention just a few. In recent work, my collaborators and I use a subset of parking functions we call *unit Fubini rankings*, and in particular their outcome map, to characterize and enumerate the Boolean intervals of rank k in the weak order poset of $\mathfrak{S}_n$ [EHKMM24]. Figure 4 illustrates our construction.

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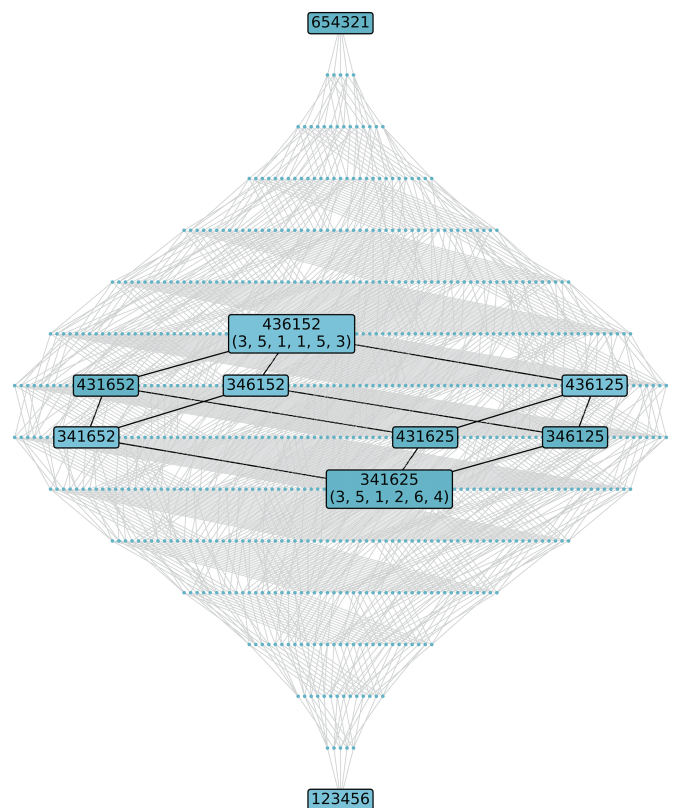
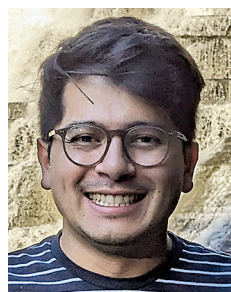


Figure 4. Weak order poset of $\mathfrak{S}_6$ with a Boolean interval of rank 3 highlighted, with minimal element 341625 and maximal element 436152 written in one-line notation. The outcome of $(3, 5, 1, 2, 6, 4)$, a parking function of length 6, is $(3, 4, 1, 6, 2, 5)$ and corresponds to the minimal element of the interval. The outcome of $(3, 5, 1, 1, 5, 3)$, another parking function of length 6, is again $(3, 4, 1, 6, 2, 5)$ and corresponds to the minimal element of the interval. In fact, $(3, 5, 1, 1, 5, 3)$ is a “unit Fubini ranking with 3 distinct ranks” and corresponds to a Boolean interval of rank $6 - 3 = 3$ (i.e., a cube), whereas $(3, 5, 1, 2, 6, 4)$ is a “unit Fubini ranking with 6 distinct ranks” and corresponds to a Boolean interval of rank $6 - 6 = 0$ (i.e., a node). Refer to Elder et al. [EHKMM24] for details.

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